

Week 6: Conversion of Numbers

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

Since computers "think" in binary but humans normally use denary (base 10), it is important to know how to convert numbers between the two systems.

Converting Base 10 → Base 2 (Division Method)

Keep dividing by 2 and record the remainders. Read the remainders from **bottom to top**.

Example 1: Convert 25 (base 10) to binary.

25 divided by 2 = 12 remainder 1
12 divided by 2 = 6 remainder 0
6 divided by 2 = 3 remainder 0
3 divided by 2 = 1 remainder 1
1 divided by 2 = 0 remainder 1

Reading remainders bottom to top: **25 (base 10) = 11001 (base 2)**

Example 2: Convert 42 (base 10) to binary.

42 divided by 2 = 21 remainder 0
21 divided by 2 = 10 remainder 1
10 divided by 2 = 5 remainder 0
5 divided by 2 = 2 remainder 1
2 divided by 2 = 1 remainder 0
1 divided by 2 = 0 remainder 1

Reading remainders bottom to top: **42 (base 10) = 101010 (base 2)**

Converting Base 2 → Base 10 (Place Value Method)

Multiply each digit by its place value (power of 2) and add.

Example 1: Convert 11001 (base 2) to base 10.

$(1 \times 16) + (1 \times 8) + (0 \times 4) + (0 \times 2) + (1 \times 1) = 16 + 8 + 0 + 0 + 1 = 25$

This confirms our earlier conversion was correct.

Example 2: Convert 101010 (base 2) to base 10.

$(1 \times 32) + (0 \times 16) + (1 \times 8) + (0 \times 4) + (1 \times 2) + (0 \times 1) = 32 + 8 + 2 = 42$

This confirms our second conversion was correct too.

Real-Life Application

When you type the letter "A" on a keyboard, the computer converts it internally to its binary code (01000001 in one common standard) so it can be stored and processed electronically. Converting between denary and binary is the basic skill behind

understanding how such codes work.

Vocabulary

Word	Meaning
Remainder	The amount left over after division
Conversion	Changing a number from one base (or form) to another

Evaluation Questions

- 1 Convert 37 to binary.
- 2 Convert 100101 (base 2) to base 10.
- 3 Convert 50 to binary, then convert your answer back to base 10 to check it.
- 4 Convert 60 to binary using the division method, showing all steps.
- 5 Convert 11111 (base 2) to base 10.
- 6 Explain why we read the remainders from bottom to top when converting to binary.
- 7 Convert 19 to binary.
- 8 Convert 110110 (base 2) to base 10.
- 9 A student converts 28 to binary and gets 11100. Check this answer by converting 11100 back to base 10.
- 10 In your own words, explain the two methods used in this topic — one for base 10 to base 2, and one for base 2 to base 10.